

# 3-1 · Understand Ratios

Connect & apply · 8 tasks · Prepares for 6.RP.A.1

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Compute  $8 \times 7$ .

\_\_\_\_\_

**2. FOUNDATION 2**

Continue the pattern: 6, 12, 18, \_\_\_\_.

\_\_\_\_\_

**3. FOUNDATION 3**

There are 5 blue counters and 7 red counters.  
How many counters in all?

\_\_\_\_\_

**4. FOUNDATION 4**

Three of eight equal slices are shaded. Write the shaded fraction.

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-1 · Understand Ratios

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Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**5. CHECK A**

$6 \times 4$ ? There are 3 dogs and 5 cats — how many animals in all?

\_\_\_\_\_

**6. CHECK B**

$7 \times 3$ ? There are 4 red marbles and 6 blue — how many marbles in all?

\_\_\_\_\_

**7. CHECK C**

$8 \times 5$ ? A team has 5 guards and 7 forwards — how many players in all?

\_\_\_\_\_

**8. TRANSFER**

In a class the ratio of students wearing sneakers to boots is 5 : 2. There are 28 students. How many wear sneakers?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-2 · Rates and Unit Rates

Connect & apply · 8 tasks · Prepares for 6.RP.A.2

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Compute  $168 \div 12$ .

\_\_\_\_\_

**2. FOUNDATION 2**

Five notebooks cost \$15. What is the cost in dollars for one notebook?

\_\_\_\_\_

**3. FOUNDATION 3**

A table gives 1 bag  $\rightarrow$  4 marbles, 2 bags  $\rightarrow$  8, 3 bags  $\rightarrow$  12. How many marbles for 5 bags?

\_\_\_\_\_

**4. FOUNDATION 4**

A cyclist rides 24 miles in 2 hours. State the unit rate with its unit.

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-2 · Rates and Unit Rates

Connect & apply · 8 tasks · Prepares for 6.RP.A.2

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**5. CHECK A**

$60 \div 4$ ? If 4 pencils cost \$2, what does 1 pencil cost?

\_\_\_\_\_

**6. CHECK B**

$72 \div 6$ ? If 6 notebooks cost \$9, what does 1 notebook cost?

\_\_\_\_\_

**7. CHECK C**

$150 \div 5$ ? If a car travels 150 miles in 5 hours, what is the speed per hour?

\_\_\_\_\_

**8. TRANSFER**

Brand A is 12 oz for \$3.00. Brand B is 20 oz for \$4.60. Which is the better buy?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-3 · Equivalent Ratios Using Tables

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Scale the pair 3 : 5 by a factor of 4.

\_\_\_\_\_

**2. FOUNDATION 2**

Compute  $9 \times 8$ .

\_\_\_\_\_

**3. FOUNDATION 3**

Fill in the numerator:  $\frac{2}{3} = \frac{\_\_}{15}$ .

\_\_\_\_\_

**4. FOUNDATION 4**

A table lists (2, 6), (4, 12), (6, 18). What pair comes next if the same step continues?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-3 · Equivalent Ratios Using Tables

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**5. CHECK A**

$2 \times 5$  and  $3 \times 5$ ? Complete:  $2 : 3 = 8 : \underline{\hspace{1cm}}$

\_\_\_\_\_

**6. CHECK B**

$4 \times 6$  and  $5 \times 6$ ? Complete:  $4 : 5 = 24 : \underline{\hspace{1cm}}$

\_\_\_\_\_

**7. CHECK C**

$3 \times 7$  and  $8 \times 7$ ? Complete:  $3 : 8 = 21 : \underline{\hspace{1cm}}$

\_\_\_\_\_

**8. TRANSFER**

A recipe uses 3 cups flour to 2 cups water. You have 8 cups of water. How much flour?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-4 · Equivalent Ratios Using Graphs

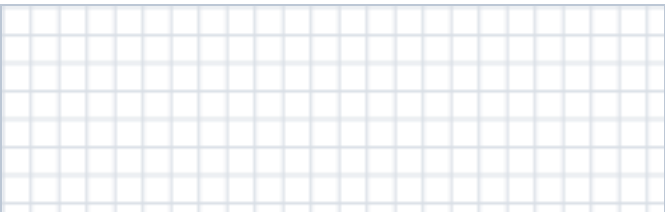
Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Describe how to plot (3, 5) from the origin.



2. FOUNDATION 2

The x-axis ticks are 0, 2, 4, 6, 8. What x-value is at the third tick to the right of 0?

\_\_\_\_\_

3. FOUNDATION 3

A point is 4 units right and 7 units up from the origin. Give its ordered pair.

\_\_\_\_\_

4. FOUNDATION 4

The rule is  $y = 3x$ . Give  $y$  when  $x = 4$ .

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-4 · Equivalent Ratios Using Graphs

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**5. CHECK A**

Start at (0, 0), move 3 right and 2 up — write the ordered pair. If each gridline is 5, where is 15?

\_\_\_\_\_

**6. CHECK B**

Start at (0, 0), move 5 right and 4 up — write the ordered pair. If each gridline is 2, where is 12?

\_\_\_\_\_

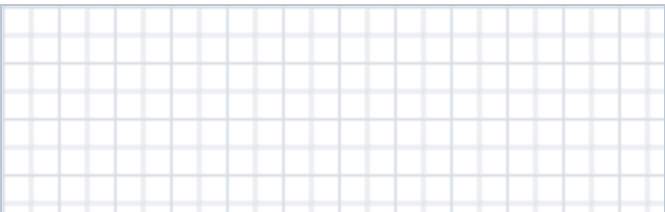
**7. CHECK C**

Start at (0, 0), move 2 right and 7 up — write the ordered pair. If each gridline is 10, where is 50?

\_\_\_\_\_

**8. TRANSFER**

A ratio graph passes through (0, 0), (2, 6), and (4, 12). What is the unit rate, and where is  $x = 5$ ?



One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-5 · Comparing Ratio Relationships

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.b

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Which is greater: 35 or 53?

\_\_\_\_\_

**2. FOUNDATION 2**

Scale 4 : 7 to an equivalent ratio with first term 20.

\_\_\_\_\_

**3. FOUNDATION 3**

What is the least common multiple of 6 and 8?

\_\_\_\_\_

**4. FOUNDATION 4**

Pack A costs \$12 for 4 pens. Pack B costs \$14 for 7 pens. Which has the lower price per pen?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-5 · Comparing Ratio Relationships

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.b

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**5. CHECK A**

Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.

\_\_\_\_\_

**6. CHECK B**

Which is greater, 18 or 15? Scale 2 : 5 so the first number is 18.

\_\_\_\_\_

**7. CHECK C**

Which is greater, 24 or 30? Scale 4 : 7 so the first number is 24.

\_\_\_\_\_

**8. TRANSFER**

Runner A covers 3 miles in 24 minutes; Runner B covers 5 miles in 45 minutes. Who is faster?

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-6 · Converting Within One System

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Compute  $72 \div 9$ .

\_\_\_\_\_

**2. FOUNDATION 2**

Use 1 foot = 12 inches. How many inches are in 3 feet?

\_\_\_\_\_

**3. FOUNDATION 3**

How many meters are in 2.4 kilometers? Use 1 km = 1000 m.

\_\_\_\_\_

**4. FOUNDATION 4**

Convert 48 inches to feet using 12 inches = 1 foot.

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-6 · Converting Within One System

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

12 × 4? 1 foot = \_\_\_\_ inches? 2.5 m = \_\_\_\_ cm?

\_\_\_\_\_

6. CHECK B

11 × 4? 1 pound = \_\_\_\_ ounces? 3.5 m = \_\_\_\_ cm?

\_\_\_\_\_

7. CHECK C

9 × 6? 1 gallon = \_\_\_\_ quarts? 0.75 km = \_\_\_\_ m?

\_\_\_\_\_

8. TRANSFER

A tank holds 3 gallons. How many cups is that? (1 gal = 4 qt, 1 qt = 4 cups)

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-7 · Converting Between Systems

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

**1. FOUNDATION 1**

Compute  $6 \times 2.5$ .

\_\_\_\_\_

**2. FOUNDATION 2**

Round 7.846 to the nearest hundredth.

\_\_\_\_\_

**3. FOUNDATION 3**

Use 1 inch = 2.54 centimeters. Convert 5 inches to centimeters.

\_\_\_\_\_

**4. FOUNDATION 4**

Use 1 yard = 3 feet and 1 foot = 12 inches. Convert 2 yards to inches.

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help

# 3-7 · Converting Between Systems

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

$4 \times 1.6$ ?  $10 \times 0.45$ ? Round 6.44 to the nearest tenth.

\_\_\_\_\_

6. CHECK B

$5 \times 2.54$ ?  $8 \times 0.45$ ? Round 12.7 to the nearest tenth.

\_\_\_\_\_

7. CHECK C

$3 \times 1.6$ ?  $20 \times 0.45$ ? Round 9.08 to the nearest tenth.

\_\_\_\_\_

8. TRANSFER

A race is 5 kilometers. About how many miles is that, if  $1 \text{ km} \approx 0.62 \text{ mi}$ ? Round to the nearest tenth.

\_\_\_\_\_

One strategy I used: \_\_\_\_\_

My next step: ☐ Explain it to someone   ☐ Try another example   ☐ Ask for help